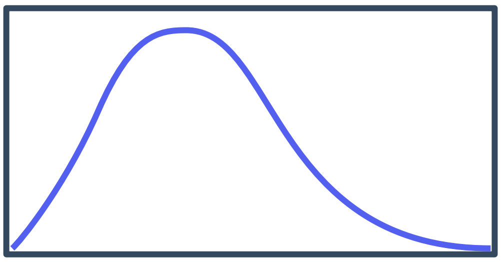
Normal distributions

1 State whether the following statements are true of the normal distribution curve:

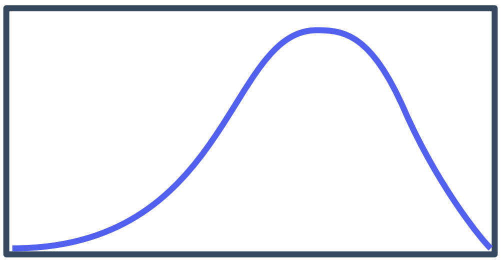
- **a** There are the same number of scores above and below the mean.
- **b** The curve is symmetrical.
- **c** The fewest scores lie around the mean.
- **d** The curve is asymmetrical.
- **e** The spread of the normal distribution changes depending on the mean.
- **f** A higher mean will result in a skewed curve.
- **g** The mean, median and mode all have the same value.
- **h** The spread of the normal distribution changes depending on the standard deviation.

2 State whether the following figures are normally distributed:

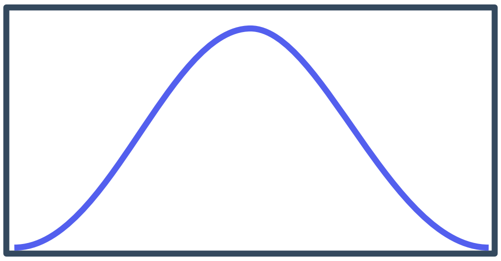
a



b

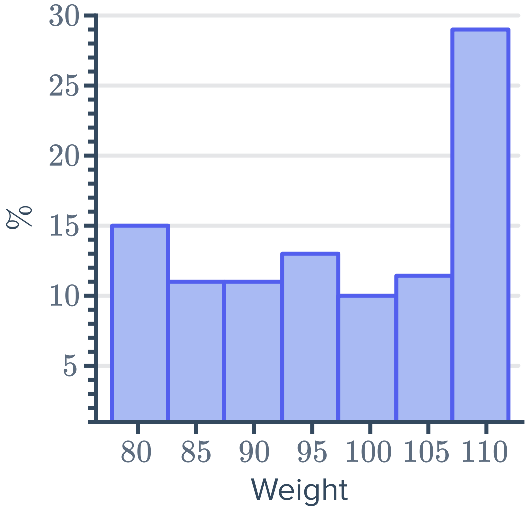


c

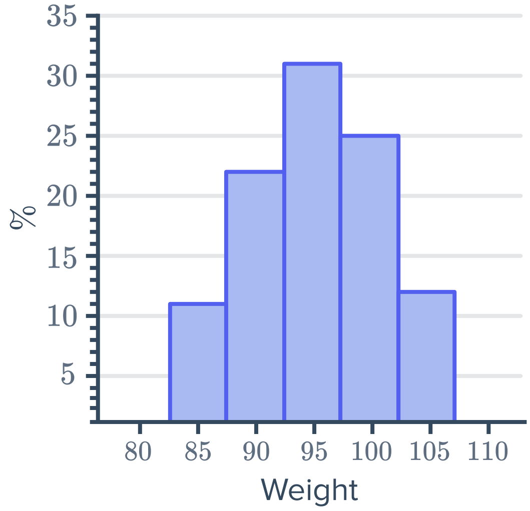


3 State whether the following histograms are approximately normally distributed:

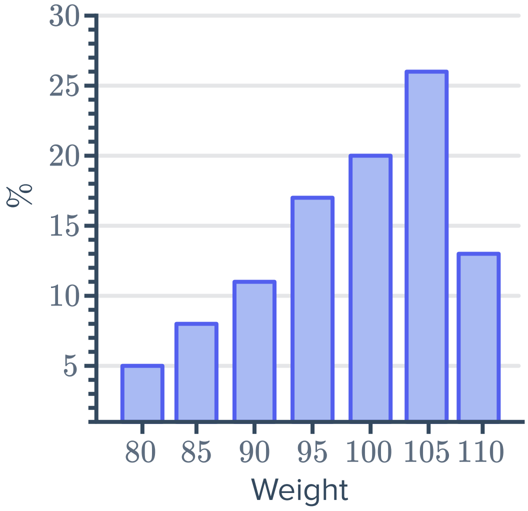
a



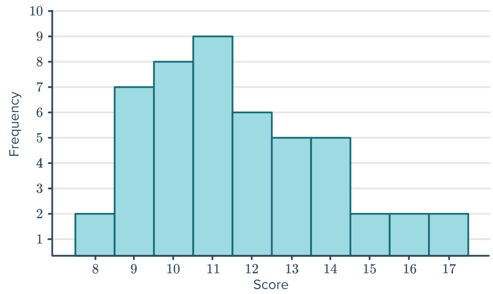
b



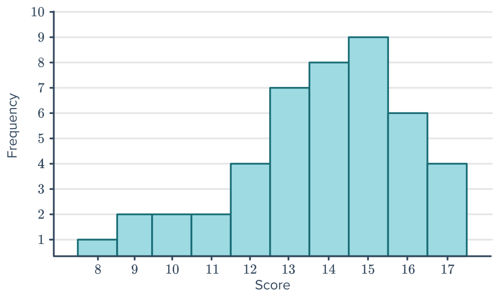
c



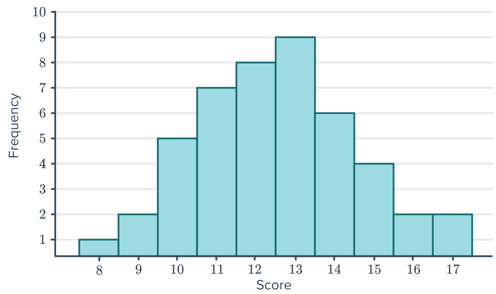
d



e

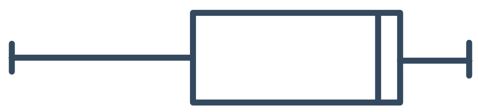


f



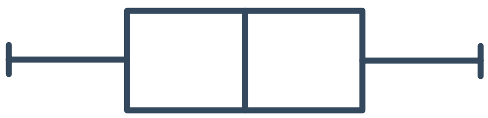
4 State whether the following boxplots could [illegible] from data which is approximately normally distributed:

a



b

c



5 Given a normal distribution, state the percentage of people that lie:

a Above the mean.

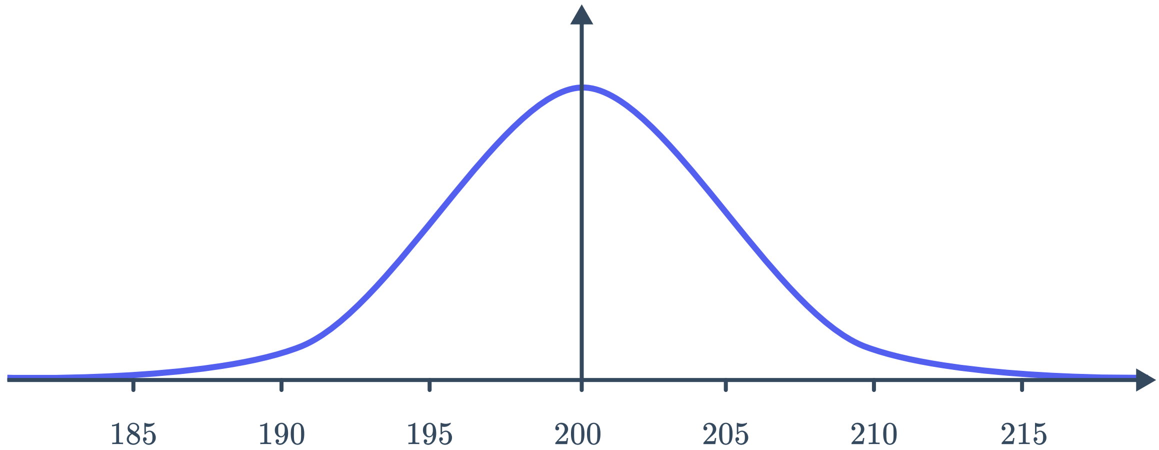
b Below the mean.

6 Consider the given normal distribution:

a State the mean.

b State the median.

c State the mode.



7 The mean of a set of scores, denoted by μ is 51 and the standard deviation, denoted by σ is 16. Find the value of the following:

a $\mu - \sigma$

b $\mu + 2\sigma$

c $\mu + 3\sigma$

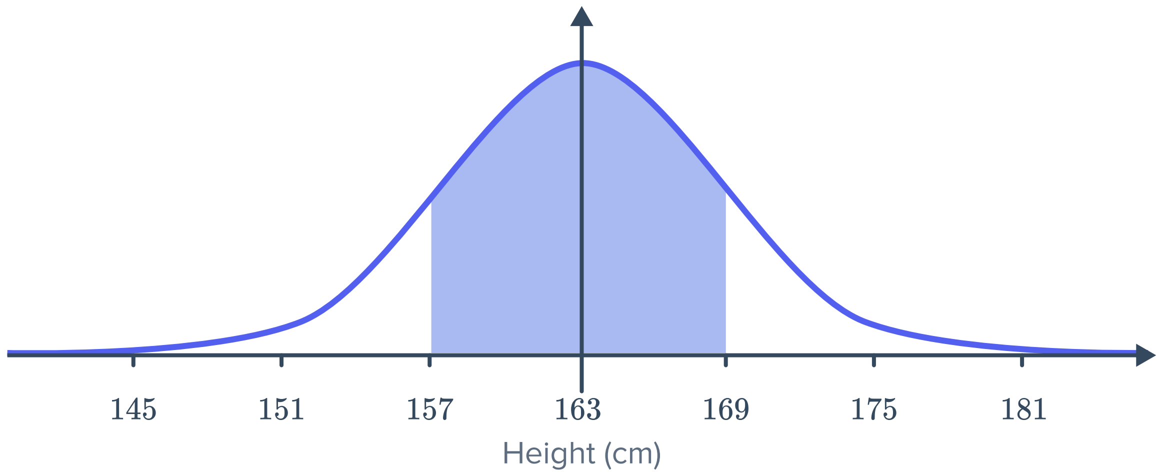
d $\mu - 2\sigma$

e $\mu + 0.5\sigma$

f $\mu - \dfrac{2\sigma}{3}$

8 A data set has a mean of 80 and a standard deviation of 3. Find the score that is 3 standard deviations above the mean.

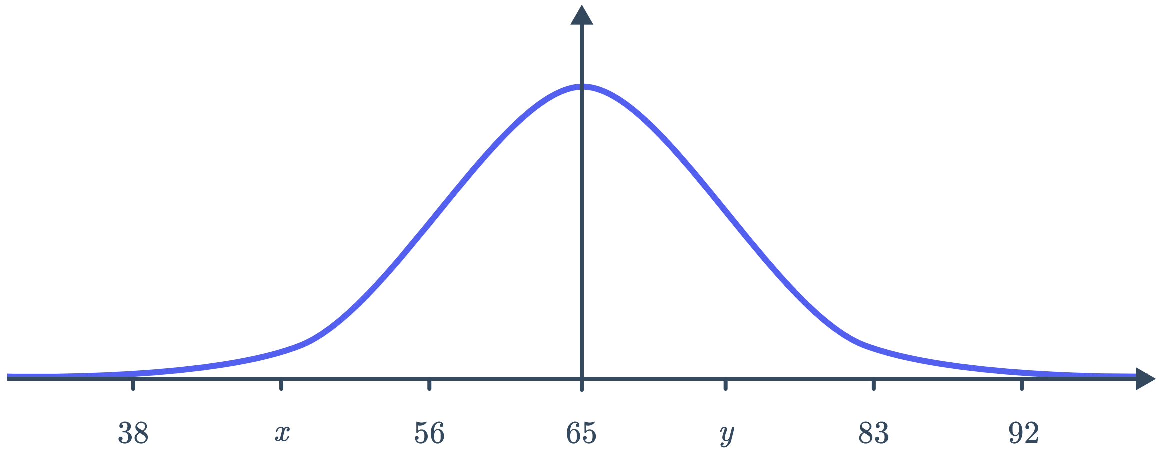
9 The following graph shows the distribution of females' heights in a population:



a If 34% of females lie between 157 cm and 163 cm, find the percentage of females that lie between 163 cm and 169 cm.

b State the value of 1 standard deviation.

10 The results from an exam were approximately normally distributed. The mean score was 65, with a standard deviation of 9. The points marked on the horizontal axis are separated by 1 standard deviation:



a State the value of x on the graph.

b State the value of y on the graph.

11 Assume the mass of sumo wrestlers is normally distributed, with a mean mass of 158 kg and a standard deviation of 10 kg.

a State the percentage of wrestler that would weigh more than 158 kg.

b How far below the mean is a sumo wrestler who weighs 148 kg.

12 In a male population, the mean height was 175 cm, with a standard deviation of 7 cm.

a State the height of Bob who is 4 standard deviations above the mean.

b State the height of John who is 2 standard deviations below the mean.

13 A sample of professional basketball players is normally distributed and gives the mean height as 199 cm with a standard deviation of 10 cm. State the height of a basketball player who is:

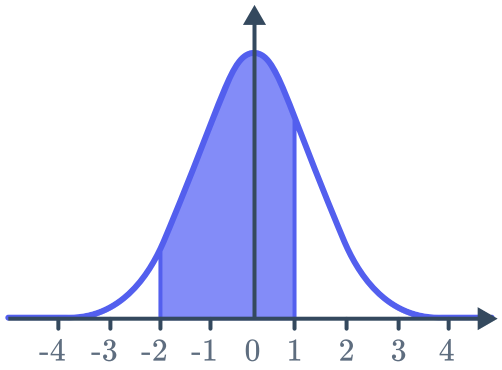
a 3 standard deviations above the mean

b 1.5 standard deviations below the mean

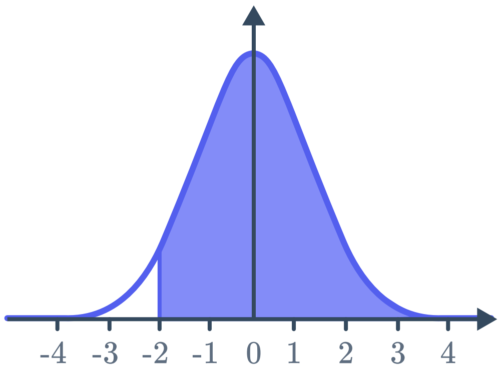
The empirical rule

14 In the following normal distributions, each unit on the horizontal axis indicates 1 standard deviation. Find the approximate percentage of scores that lie in the shaded region:

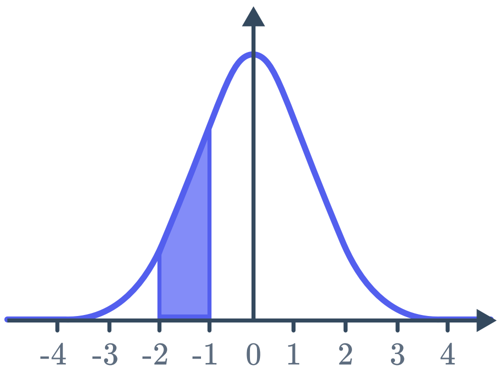
a



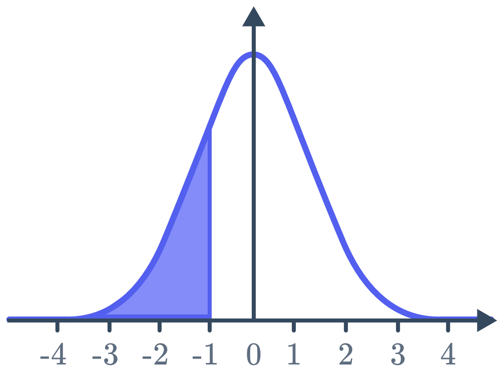
b



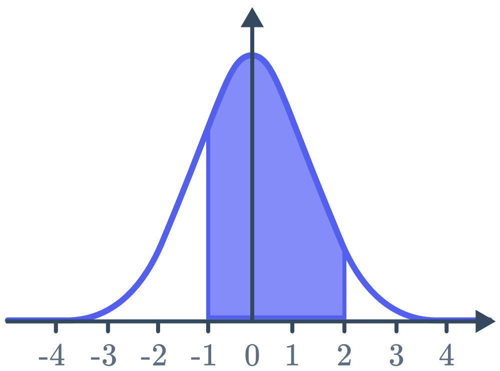
c



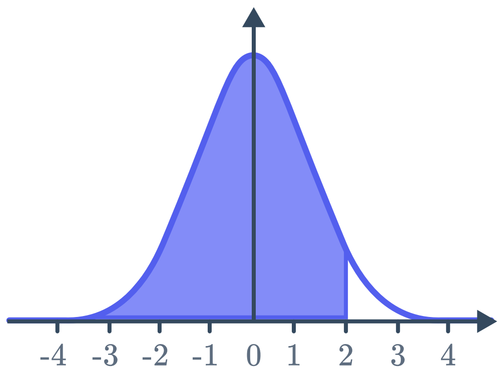
d



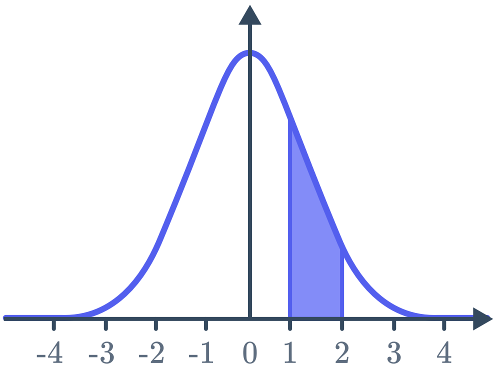
e



f



g



15 In a normal distribution, state the approximate percentage of the scores that lie within:

a 1 standard deviation of the mean.

b 2 standard deviations of the mean.

c 3 standard deviations of the mean.

16 In a normal distribution, state the approximate percentage of scores that lie between the mean and:

a 1 standard deviation above the mean.

b 2 standard deviations below the mean.

c 3 standard deviations below the mean.

17 In a normal distribution, state the approximate percentage of scores that lie between:

a 1 standard deviation above and 2 standard deviations below the mean.

b 1 standard deviation below and 3 standard deviations above the mean.

c 2 standard deviations below and 3 standard deviations above the mean.

18 In a normal distribution, the percentages of scores that lie within x standard deviation(s) of the mean are given below. Find the value of x for each percentage:

a Approximately 68% of scores lie within x standard deviation(s) of the mean.

b Approximately 95% of scores lie within x standard deviation(s) of the mean.

c Approximately 99.7% of scores lie within x standard deviation(s) of the mean.

19 In a normal distribution, the percentages of scores that lie between the mean and x standard deviation(s) above the mean are given below. Find the value of x for each percentage.

a) Approximately 34% of scores lie between the mean and x standard deviation(s) above the mean.

b) Approximately 47.5% of scores lie between the mean and x standard deviation(s) above the mean.

c) Approximately 49.85% of scores lie between the mean and x standard deviation(s) above the mean.

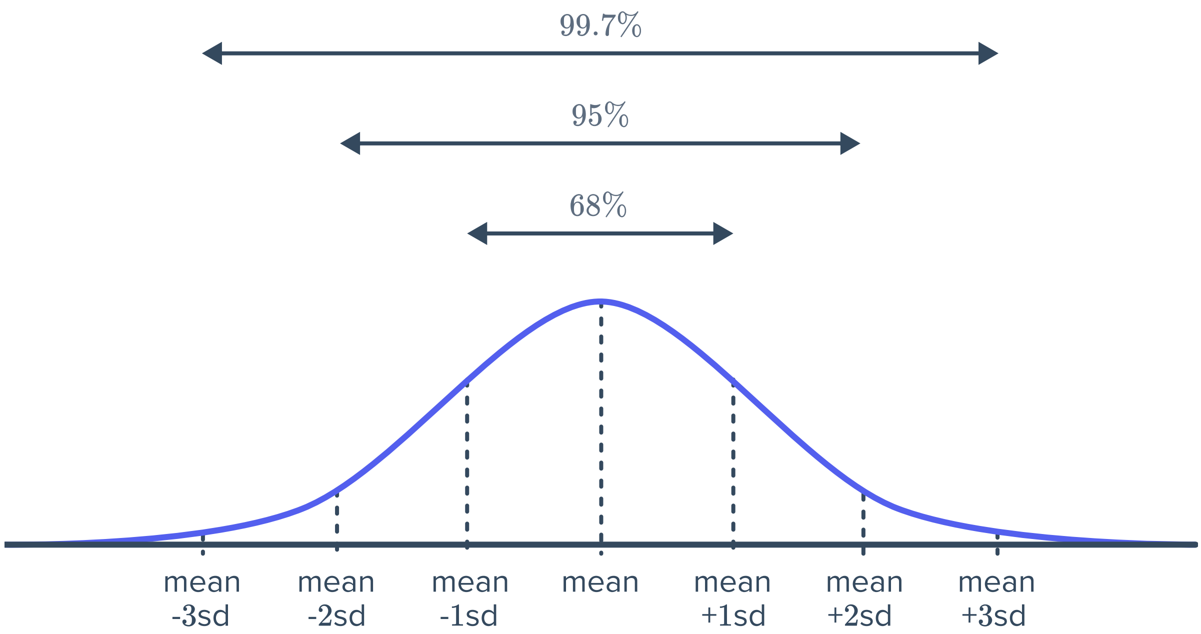
20 The grades in a test are approximately normally distributed. The mean mark is 60 with a standard deviation of 2. The approximate percentages of results that lie symmetrically about the mean are given below. Find the two scores that each of the following percentages lie between:

a) Approximately 68%.

b) Approximately 95%.

c) Approximately 99.7%.

21 The figure shows the approximate percentage of scores lying within various standard deviations from the mean of a normal distribution:



The heights of 600 boys are found to approximately follow such a distribution, with a mean height of 145 cm and a standard deviation of 20 cm. Find the number of boys with heights between:

a) 125 cm and 165 cm.

b) 105 cm and 185 cm .

c) 85 cm and 205 cm .

d) 145 cm and 165 cm .

e) 165 cm and 185 cm .

22 The heights of 400 netball players were measured and found to fit a normal distribution. The mean height is 149 cm and the standard deviation is 11. Find the number of players, that would be expected to have a height between:

- **a** 138 cm and 160 cm.
- **b** 127 cm and 171 cm.
- **c** 116 cm and 182 cm.
- **d** 149 cm and 160 cm.
- **e** 160 cm and 171 cm.

23 The number of biscuits packaged in biscuit boxes is approximately normally distributed with mean 38 and standard deviation 5. If 4000 boxes of biscuits are produced, find the approximate number of boxes with more than 33 biscuits.

24 The heights of players in a soccer club are approximately normally distributed, with mean height 1.76 m and standard deviation 5 cm. If 700 players are chosen at random, find the approximate number of players who are taller than 1.66 m.

25 The times that professional divers can hold their breath are approximately normally distributed with mean 106 seconds and standard deviation 8 seconds. If 700 professional divers are selected at random, find the approximate number of divers that would be able to hold their breath for longer than 82 seconds.

26 A set of scores is approximately normally distributed, where the mean score is 92 and standard deviation is 20. Using the empirical rule, find the percentage of scores between:

- **a** 72 and 112.
- **b** 52 and 132.
- **c** 32 and 152.
- **d** 92 and 112.
- **e** 112 and 132.

27 For a set of scores, it was found that the mean score was 61 and the standard deviation was 11. If the distribution of scores is approximately normal, find the percentage of scores between:

- **a** 50 and 72.
- **b** 39 and 83.
- **c** 28 and 94.
- **d** 61 and 72.
- **e** 72 and 83.

28 The marks in a class were approximately normally distributed. If the mean was 47 with a standard deviation of 6, Find the percentage of students who achieved:

a A mark above the average.

b A mark above 41.

c A mark above 53.

d A mark below 35.

e A mark above 29.

f A mark between 29 and 47.

g A mark between 53 and 59.

29 The times that a class of students spent talking or texting on their phones on a particular weekend is approximately normally distributed with mean time 173 minutes and standard deviation 4 minutes. Find the percentage of students that used their phones for between 165 and 181 minutes on the weekend.

30 The height of sunflowers is approximately normally distributed, with a mean height of $1.6\ \mathrm{m}$ and a standard deviation of $5\ \mathrm{cm}$.

a Find the percentage of sunflowers between $1.5\ \mathrm{m}$ and $1.75\ \mathrm{m}$ tall.

b Find the percentage of sunflowers between $1.55\ \mathrm{m}$ and $1.75\ \mathrm{m}$ tall.

c If there are 3000 sunflowers, find the number of sunflowers taller than $1.5\ \mathrm{m}$.

31 The exams scores of students are approximately normally distributed with a mean score of 63 and a standard deviation of 8. Use the empirical rule to find:

a The approximate percentage of students who scored between 39 and 79.

b The approximate number of students who passed if there are 450 students in the class and if the passing score is 39.

32 The operating times of phone batteries are approximately normally distributed with mean 34 hours and a standard deviation of 4 hours.

a Find the percentage of batteries that last between 22 and 42 hours.

b Find the percentage of batteries that last between 30 hours and 42 hours.

c Any battery that lasts less than 22 hours is deemed faulty. If a company manufactured $51\,000$ batteries, find the approximate number of batteries they would be able to sell.

33 The number of biscuits in a box is approximately normally distributed with mean 30 and standard deviation of 3.

a Approximately 81.5% of the scores lie between 2 standard deviations below and x standard deviation(s) above the mean. Find the value of x.

b Find the range of the numbers of biscuits in 81.5% of the boxes.

34 The weights of an adult harp seals are approximately normally distributed with mean 144 kg and standard deviation of 6 kg.

a Approximately 83.85% of adult harp seals lie between 1 standard deviation below and x standard deviation(s) above the mean. Find the value of x.

b Hence, find the range of the weight of 83.85% adult males, in kilograms.

35 The times for runners to complete a 100 m race is approximately normally distributed with mean 14 seconds and standard deviation of 1.9 seconds.

a Approximately 97.35% of people lie between 3 standard deviations below and x standard deviation(s) above the mean. Find the value of x.

b Hence, find the range of time in seconds for which 97.35% of runners completed the race. Round your answer to one decimal place.